

Source Code & Output

import this library

1.scikit-learn

2.seaborn

3.numpy

4.pandas

5.matplotlib

PC File Edit View Navigate Code Refactor Run Tools VCS Window Help final project - project2.py

final project > project2.py

Project

- final project [ex] C:\Users\Diya\OneDrive\Desktop\4 sem
- Mall_Customers.csv
- project2.py
- External Libraries
- Scratches and Consoles

project2.py

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 df = pd.read_csv("Mall_Customers.csv")
7 print(df)
8
```

Notifications

C:\Users\Diya\AppData\Local\JetBrains\PyCharmCE2022.1\demo\PyCharmLearningProject\venv\Scripts\python.exe "C:/Users/Diya/OneDrive/Desktop/4 sem/foml/final pr

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
..	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

[200 rows x 5 columns]

Process finished with exit code 0

Structure Bookmarks

Version Control Run TODO Problems Terminal Python Packages Python Console Services

PEP 8: W292 no newline at end of file 7:10 CRLF UTF-8 4 spaces Python 3.10 (PyCharmLearningProject)

The screenshot shows an IDE window titled "final project - project2.py". The left sidebar displays a project tree with "final project [ex]" containing "Mall_Customers.csv" and "project2.py". The main editor shows the following Python code:

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 #import seaborn as sns
5
6 df = pd.read_csv("Mall_Customers.csv")
7 print(df)
8
9 df.head()
10 df.info()
11 df.describe()
```

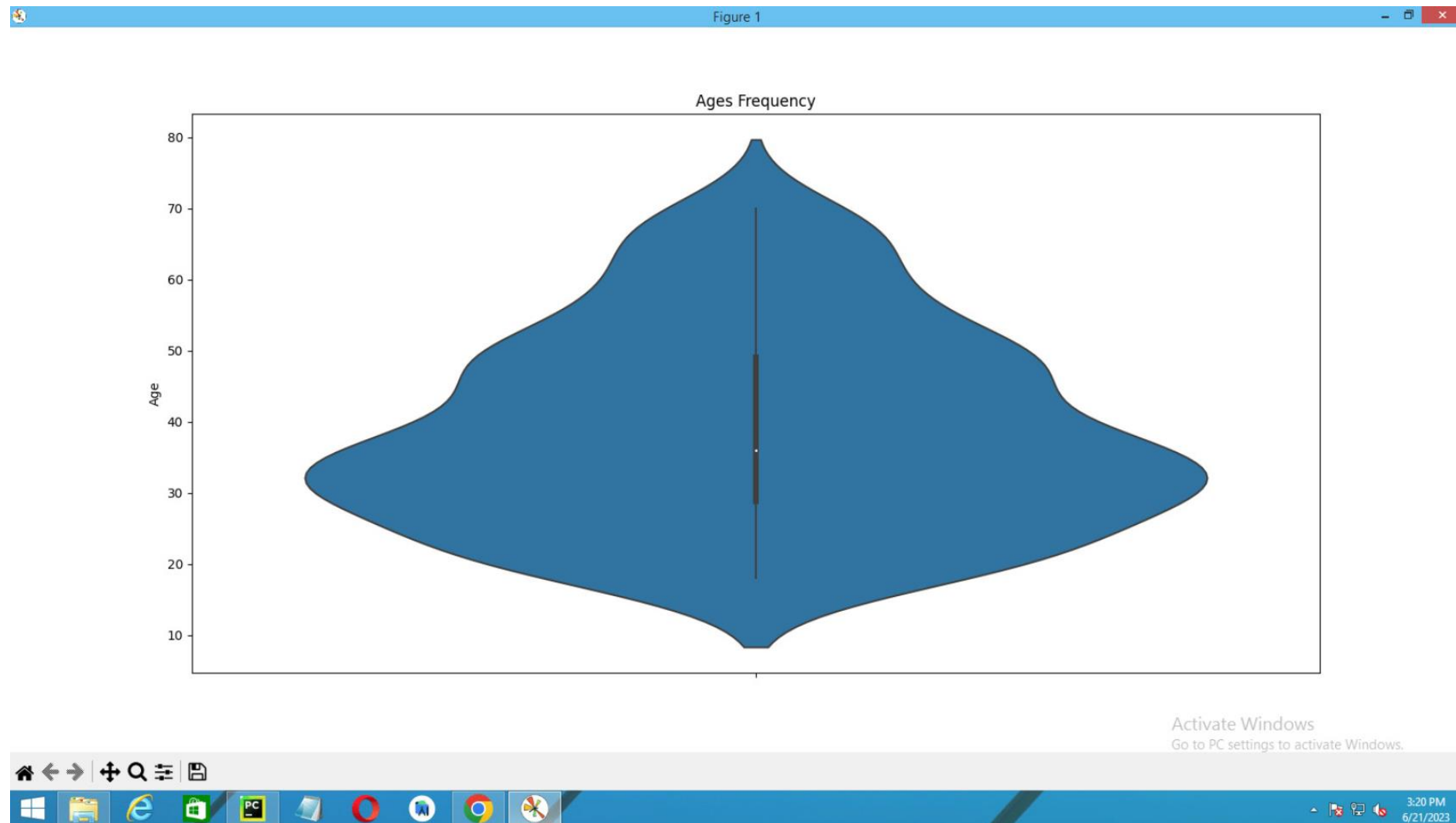
Line 11 is currently selected. The right sidebar shows a "Notifications" panel with two yellow warning icons.

The screenshot shows the console output of the Python script. The output displays the DataFrame's class, index range, column information, dtypes, and memory usage. The console also shows the process finished with exit code 0.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            200 non-null   int64
1   Gender                200 non-null   object
2   Age                   200 non-null   int64
3   Annual Income (k$)    200 non-null   int64
4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

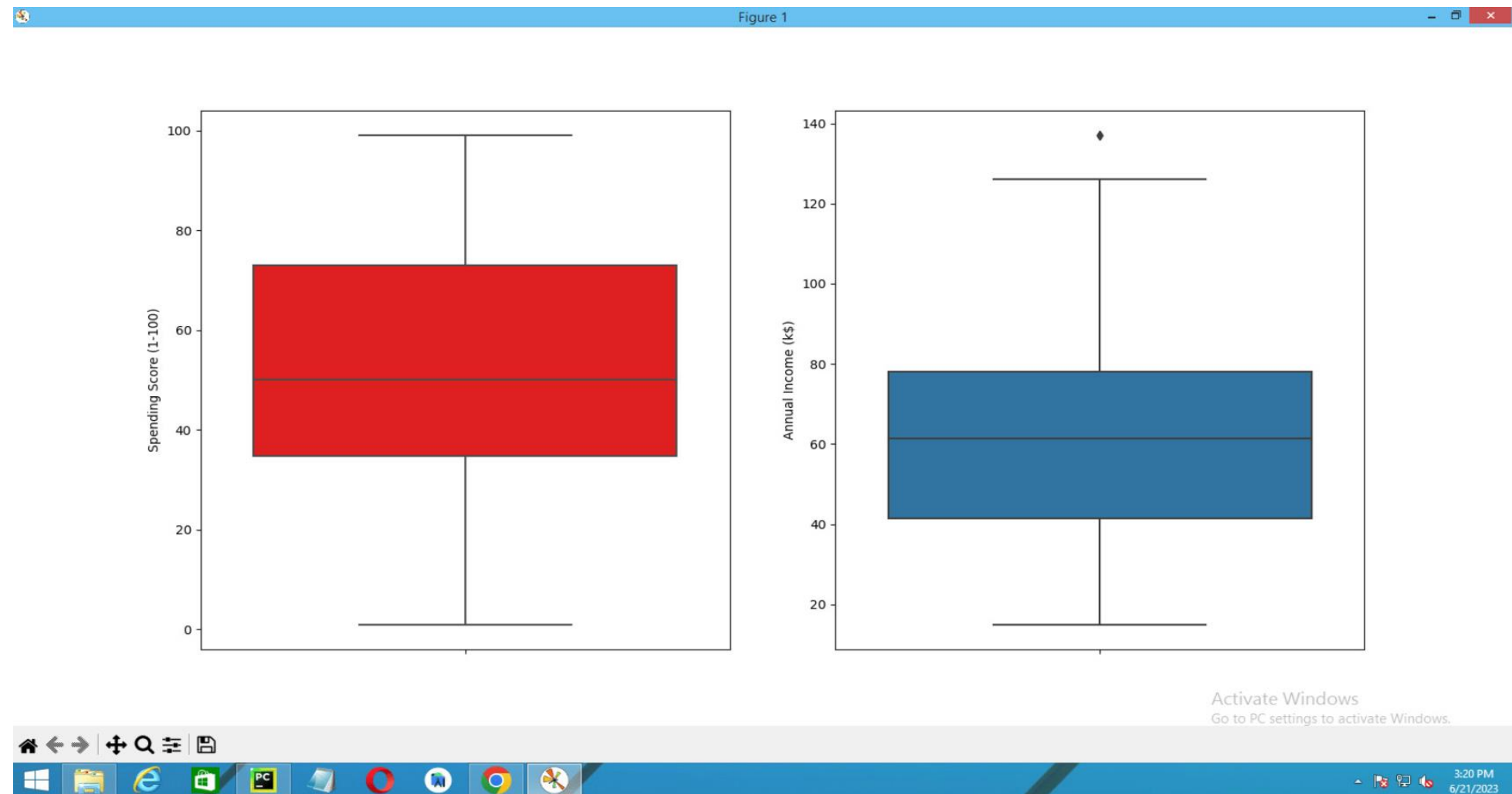
Process finished with exit code 0
```

```
13
14 df.drop(["CustomerID"], axis=1, inplace=True)
15
16 plt.figure(figsize=(10, 6))
17 plt.title("Ages Frequency")
18 sns.axes_style("dark")
19 sns.violinplot(y=df["Age"])
20 plt.show()
21
```

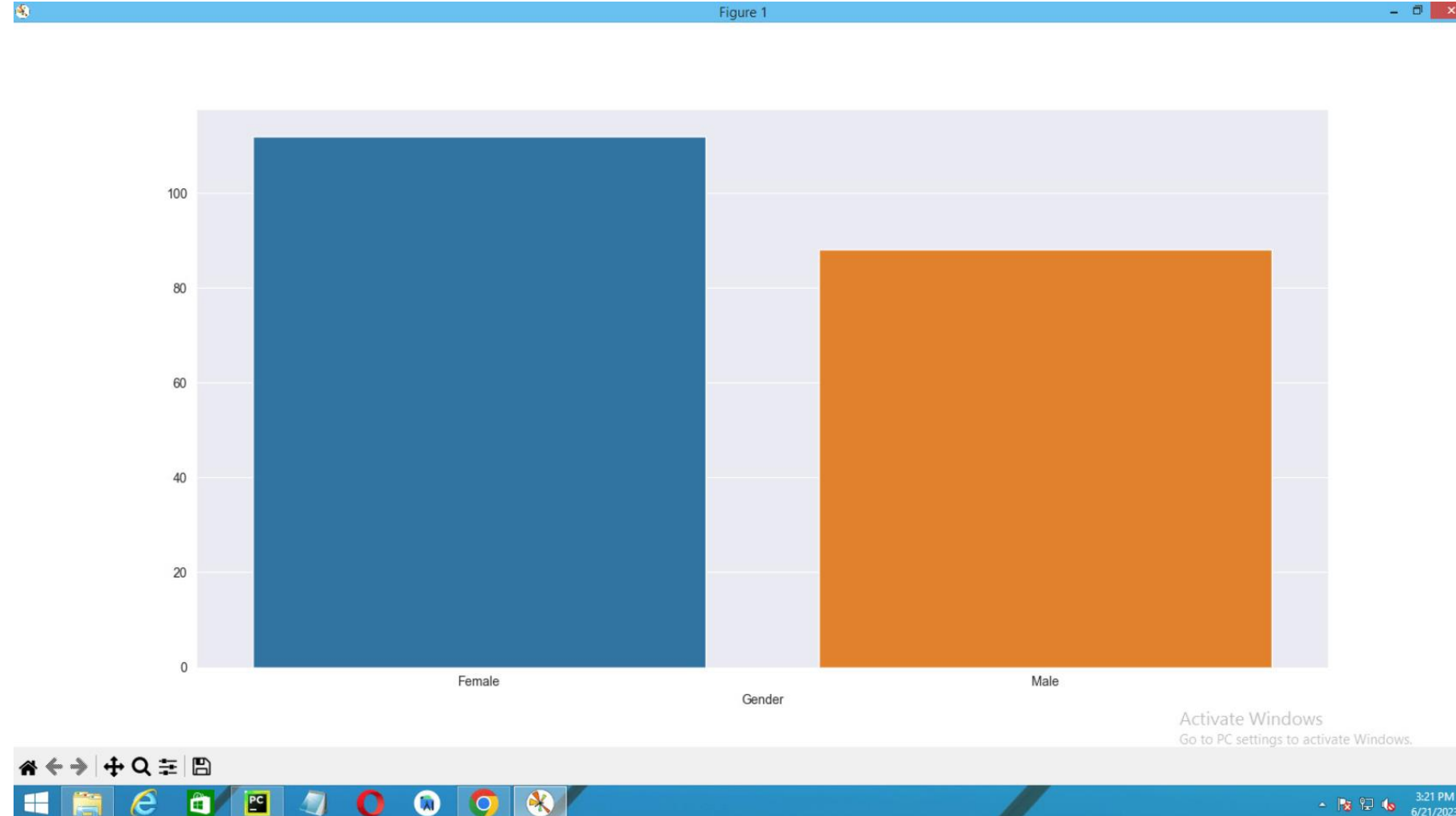


```
Mail_Customers.csv
project2.py
> External Libraries
> Scratches and Consoles

16 plt.figure(figsize=(10, 6))
17 plt.title("Ages Frequency")
18 sns.axes_style("dark")
19 sns.violinplot(y=df["Age"])
20 plt.show()
21
22 plt.figure(figsize=(15, 6))
23 plt.subplot(1, 2, 1)
24 sns.boxplot(y=df["Spending Score (1-100)"], color="red")
25 plt.subplot(1, 2, 2)
26 sns.boxplot(y=df["Annual Income (k$)"])
27 plt.show()
```



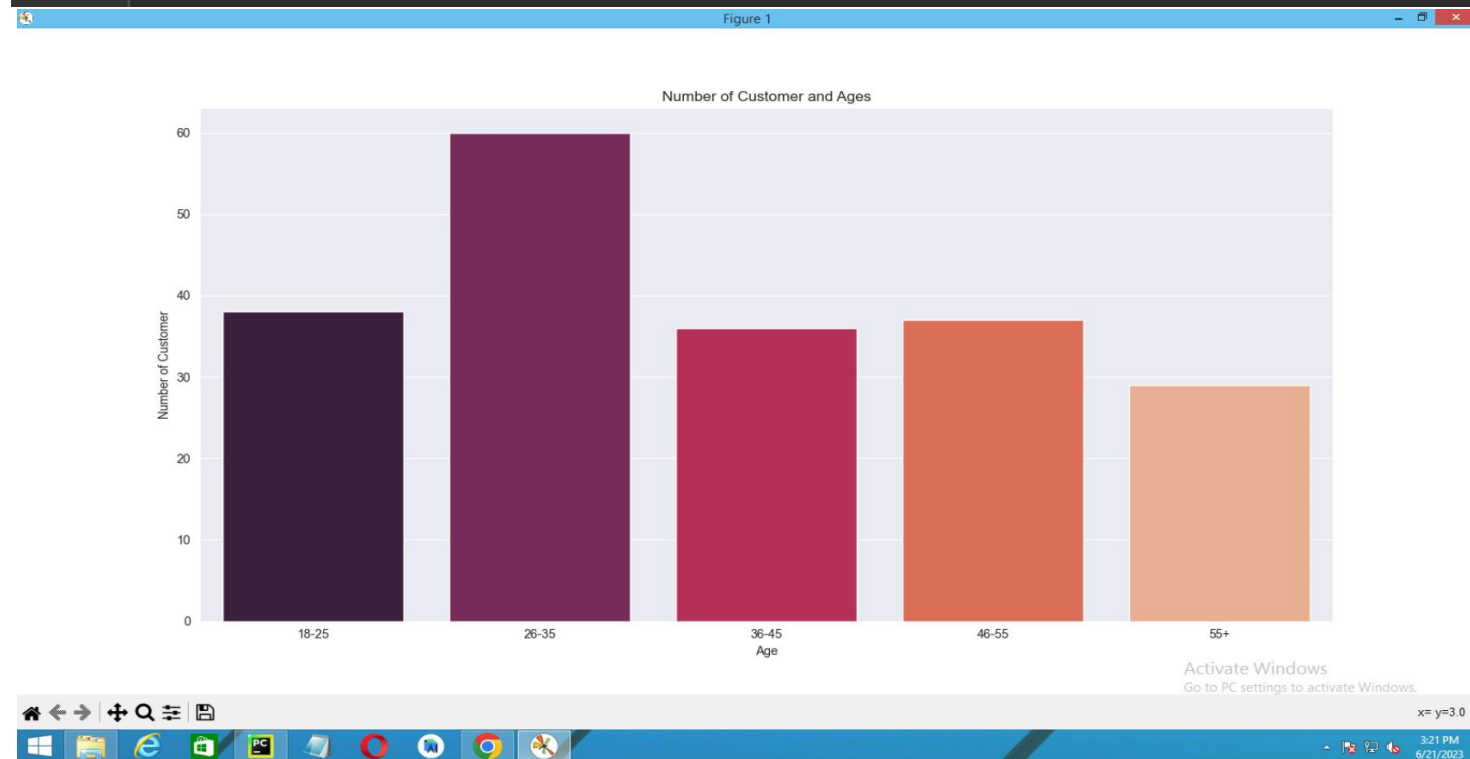
```
28
29 genders = df.Gender.value_counts()
30 sns.set_style("darkgrid")
31 plt.figure(figsize=(10, 4))
32 sns.barplot(x=genders.index, y=genders.values)
33 plt.show()
34
```



```

34
35 age18_25 = df.Age[(df.Age <= 25) & (df.Age >= 18)]
36 age26_35 = df.Age[(df.Age <= 35) & (df.Age >= 26)]
37 age36_45 = df.Age[(df.Age <= 45) & (df.Age >= 36)]
38 age46_55 = df.Age[(df.Age <= 55) & (df.Age >= 46)]
39 age55above = df.Age[df.Age >= 56]
40
41 x = ["18-25", "26-35", "36-45", "46-55", "55+"]
42 y = [len(age18_25.values), len(age26_35.values), len(age36_45.values), len(age46_55.values), len(age55above.values)]
43
44 plt.figure(figsize=(15, 6))
45 sns.barplot(x=x, y=y, palette="rocket")
46 plt.title("Number of Customer and Ages")
47 plt.xlabel("Age")
48 plt.ylabel("Number of Customer")
49 plt.show()
50 |

```

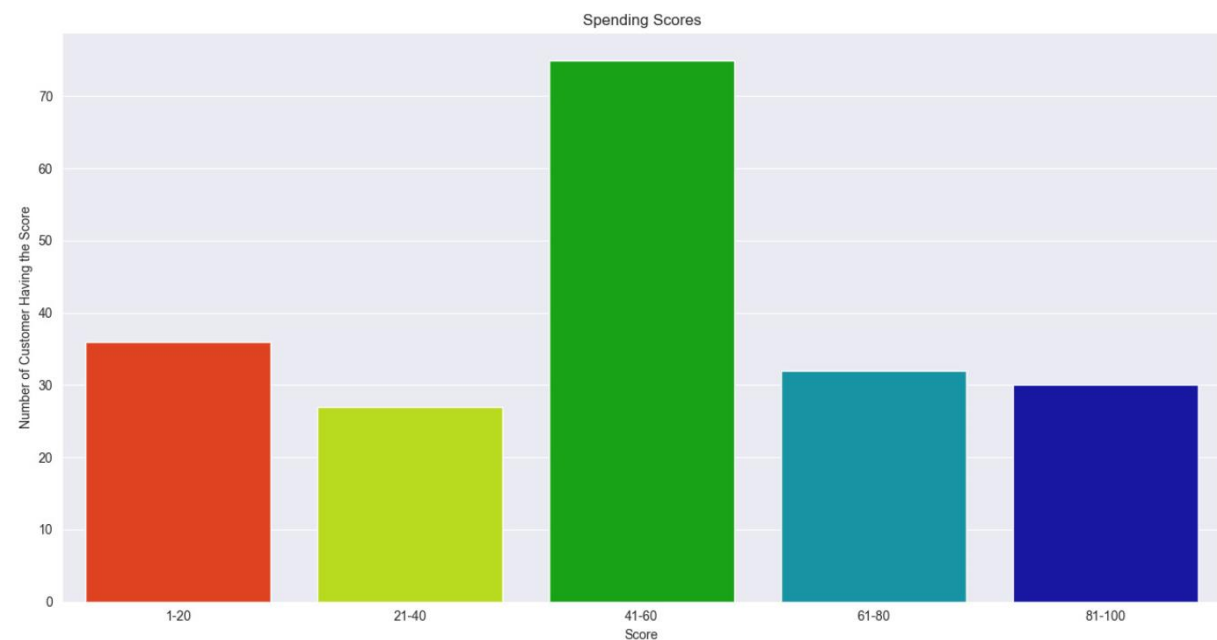


```

51 ss1_20 = df["Spending Score (1-100)"][(df["Spending Score (1-100)"] >= 1) & (df["Spending Score (1-100)"] <= 20)]
52 ss21_40 = df["Spending Score (1-100)"][(df["Spending Score (1-100)"] >= 21) & (df["Spending Score (1-100)"] <= 40)]
53 ss41_60 = df["Spending Score (1-100)"][(df["Spending Score (1-100)"] >= 41) & (df["Spending Score (1-100)"] <= 60)]
54 ss61_80 = df["Spending Score (1-100)"][(df["Spending Score (1-100)"] >= 61) & (df["Spending Score (1-100)"] <= 80)]
55 ss81_100 = df["Spending Score (1-100)"][(df["Spending Score (1-100)"] >= 81) & (df["Spending Score (1-100)"] <= 100)]
56
57 ssx = ["1-20", "21-40", "41-60", "61-80", "81-100"]
58 ssy = [len(ss1_20.values), len(ss21_40.values), len(ss41_60.values), len(ss61_80.values), len(ss81_100.values)]
59
60 plt.figure(figsize=(15, 6))
61 sns.barplot(x=ssx, y=ssy, palette="nipy_spectral_r")
62 plt.title("Spending Scores")
63 plt.xlabel("Score")
64 plt.ylabel("Number of Customer Having the Score")
65 plt.show()
66

```

Figure 1



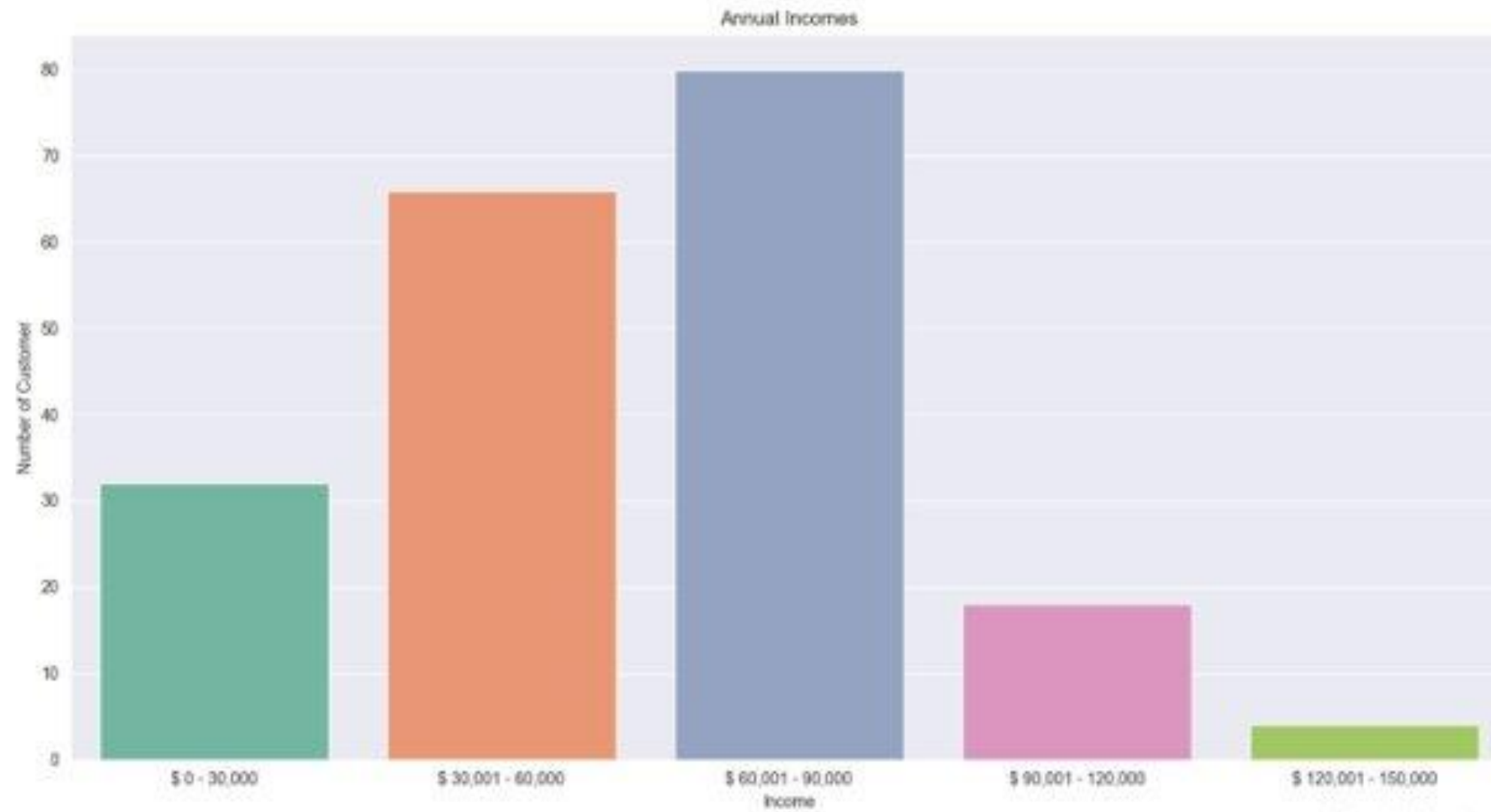
Activate Windows
Go to PC settings to activate Windows.

```
ai0_30 = df["Annual Income (k$)"][(df["Annual Income (k$)"] >= 0) & (df["Annual Income (k$)"] <= 30)]
ai31_60 = df["Annual Income (k$)"][(df["Annual Income (k$)"] >= 31) & (df["Annual Income (k$)"] <= 60)]
ai61_90 = df["Annual Income (k$)"][(df["Annual Income (k$)"] >= 61) & (df["Annual Income (k$)"] <= 90)]
ai91_120 = df["Annual Income (k$)"][(df["Annual Income (k$)"] >= 91) & (df["Annual Income (k$)"] <= 120)]
ai121_150 = df["Annual Income (k$)"][(df["Annual Income (k$)"] >= 121) & (df["Annual Income (k$)"] <= 150)]

aix = ["$ 0 - 30,000", "$ 30,001 - 60,000", "$ 60,001 - 90,000", "$ 90,001 - 120,000", "$ 120,001 - 150,000"]
aiy = [len(ai0_30.values), len(ai31_60.values), len(ai61_90.values), len(ai91_120.values), len(ai121_150.values)]

plt.figure(figsize=(15, 6))
sns.barplot(x=aix, y=aiy, palette="Set2")
plt.title("Annual Incomes")
plt.xlabel("Income")
plt.ylabel("Number of Customer")
plt.show()
```

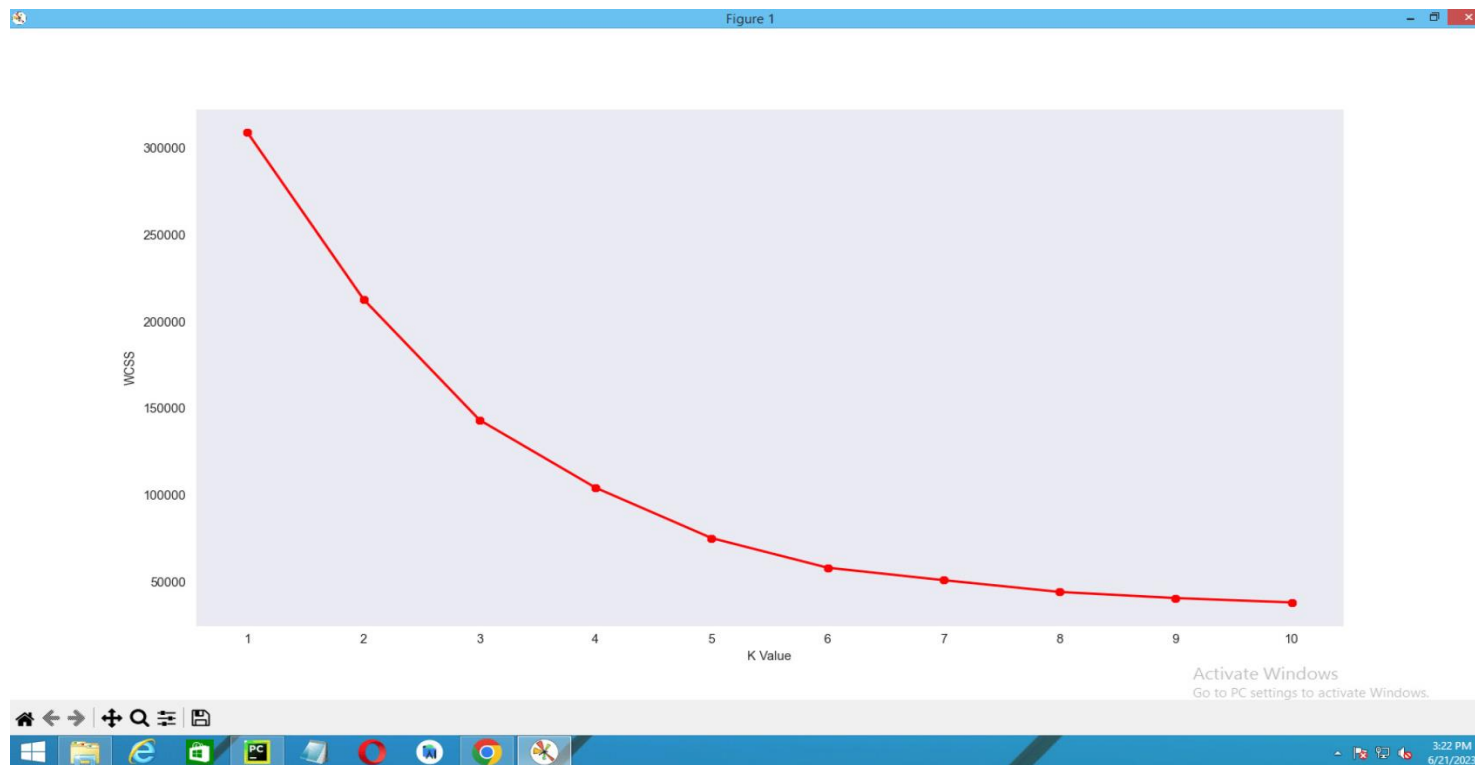
Figure 1



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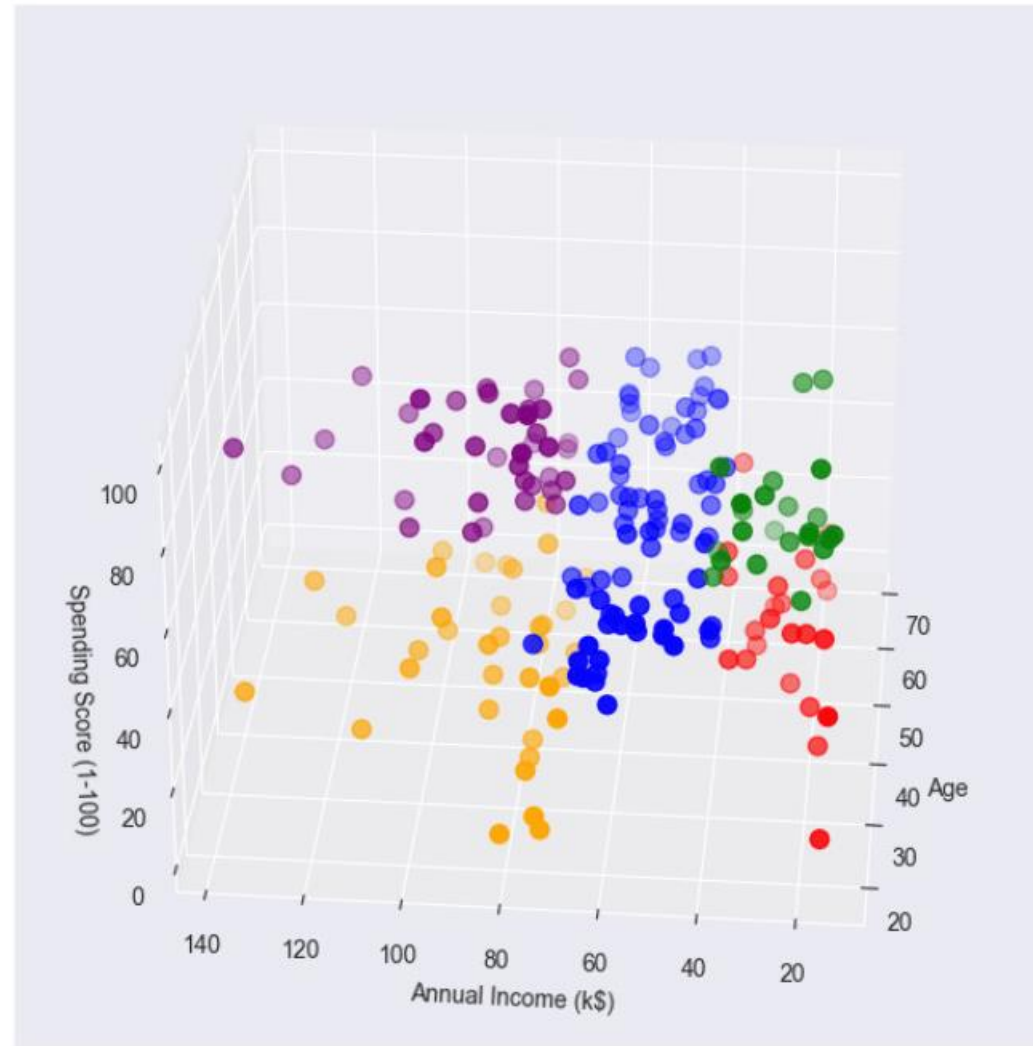
```
85
86 from sklearn.cluster import KMeans
87 wcss = []
88 for k in range(1,11):
89     kmeans = KMeans(n_clusters=k, init="k-means++")
90     kmeans.fit(df.iloc[:,1:])
91     wcss.append(kmeans.inertia_)
92 plt.figure(figsize=(12,6))
93 plt.grid()
94 plt.plot(range(1,11),wcss, linewidth=2, color="red", marker="8")
95 plt.xlabel("K Value")
96 plt.xticks(np.arange(1,11,1))
97 plt.ylabel("WCSS")
98 plt.show()
99
```



```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

fig = plt.figure(figsize=(20))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(df.Age[df.label == 0], df["Annual Income (k$)"][df.label == 0], df["Spending Score (1-100)"][df.label == 0],
          c='blue', s=60)
ax.scatter(df.Age[df.label == 1], df["Annual Income (k$)"][df.label == 1], df["Spending Score (1-100)"][df.label == 1],
          c='red', s=60)
ax.scatter(df.Age[df.label == 2], df["Annual Income (k$)"][df.label == 2], df["Spending Score (1-100)"][df.label == 2],
          c='green', s=60)
ax.scatter(df.Age[df.label == 3], df["Annual Income (k$)"][df.label == 3], df["Spending Score (1-100)"][df.label == 3],
          c='orange', s=60)
ax.scatter(df.Age[df.label == 4], df["Annual Income (k$)"][df.label == 4], df["Spending Score (1-100)"][df.label == 4],
          c='purple', s=60)
ax.view_init(30, 185)
plt.xlabel("Age")
plt.ylabel("Annual Income (k$)")
ax.set_zlabel('Spending Score (1-100)')
plt.show()
```

Figure 1



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Go to PC settings to activate Windows.

